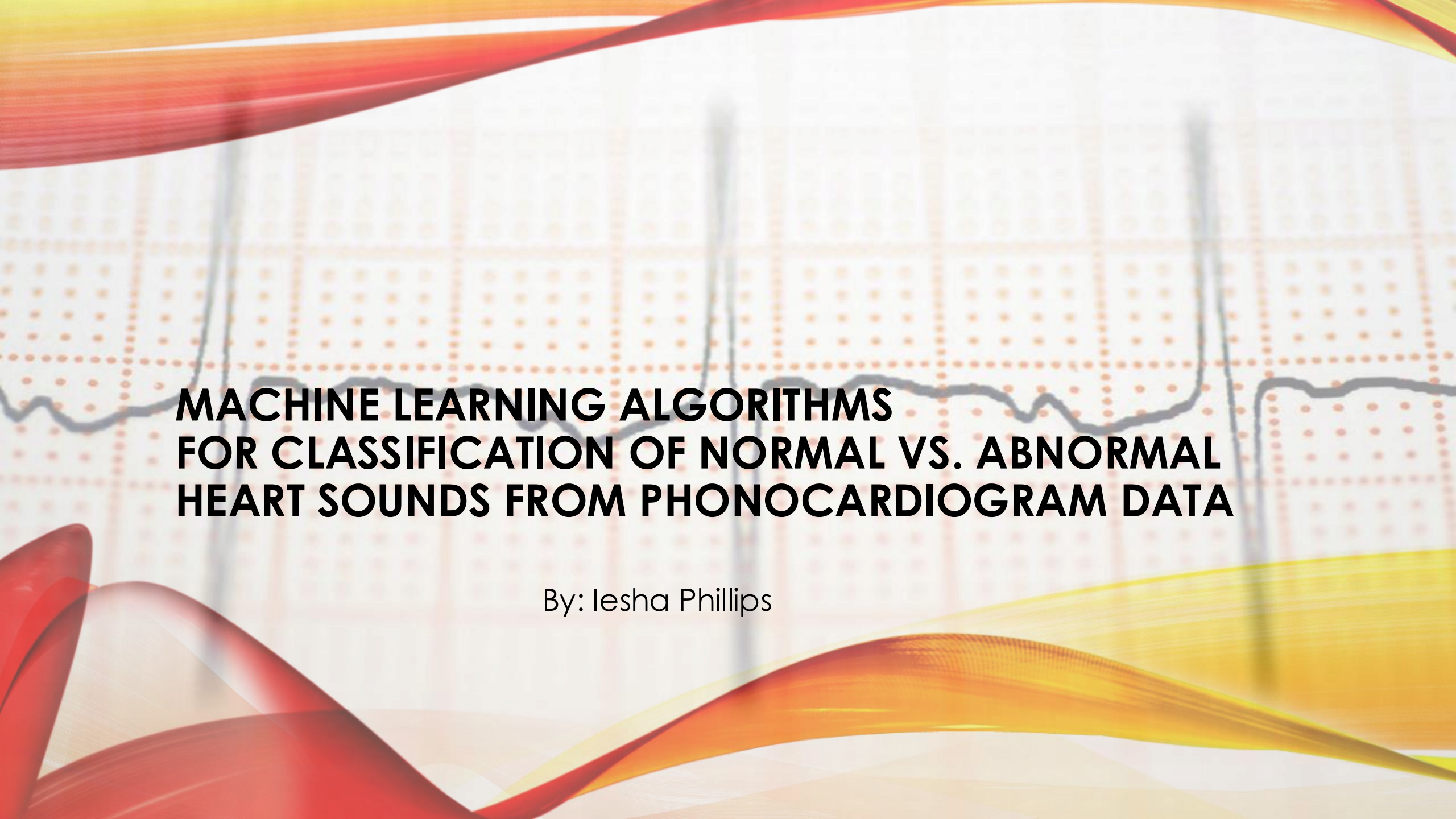




MACHINE LEARNING ALGORITHMS FOR CLASSIFICATION OF NORMAL VS. ABNORMAL HEART SOUNDS FROM PHONOCARDIOGRAM DATA

By: Iesha Phillips

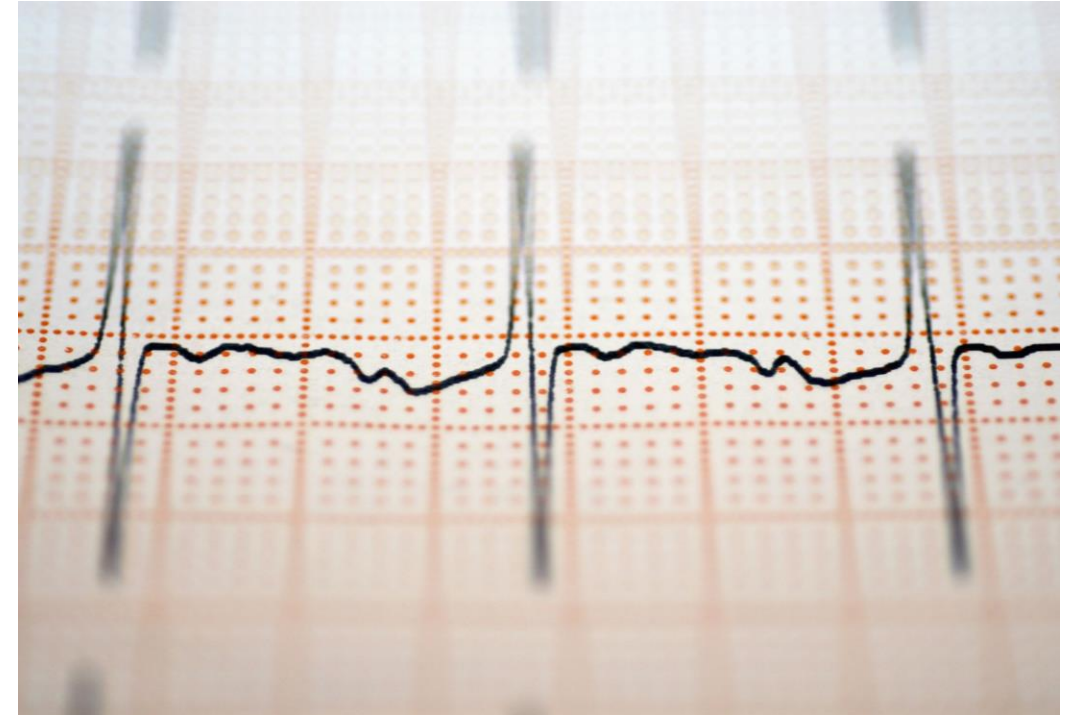
Problem Statement

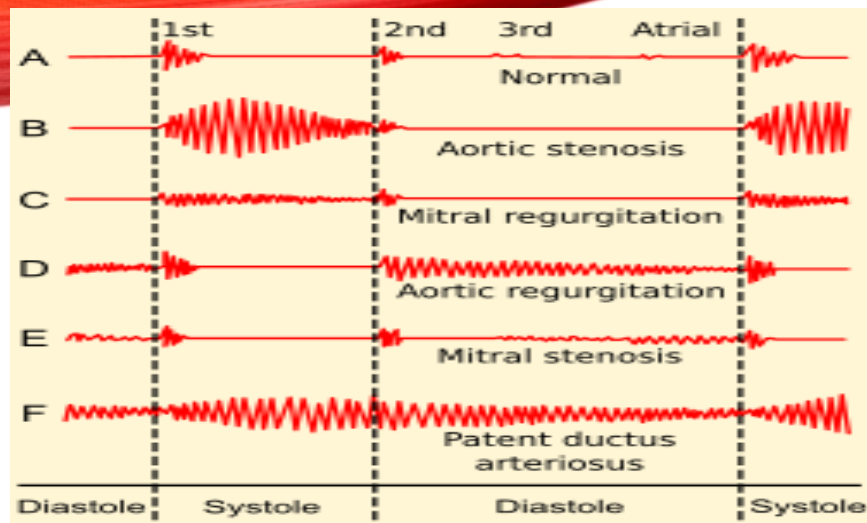
- Cardiovascular disease is a leading cause of death worldwide.
- Detecting abnormal heart sounds through auscultation is challenging due to noise, patient variability, and subtle murmur characteristics.
- Heart murmur detection often relies on clinician experience and subjective interpretation.
- These limitations can lead to missed murmurs and delayed diagnosis

Project Overview

- The goal of this project is to improve automated heart murmur detection using machine learning.
- Phonocardiogram (PCG) recordings are analyzed to classify patients with murmurs vs. no murmurs heart sounds.
- Two machine learning approaches are explored:
 1. A one-dimensional CNN operating directly on time-domain waveforms
 2. A two-dimensional CNN trained on time-frequency spectrograms

INTRODUCTION





BACKGROUND & DATABASE



- Heart sounds consist of S1 and S2 with additional components that may indicate pathology
- Heart murmurs arise from turbulent blood flow caused by valve stenosis, regurgitation, or structural defects
- Murmur characteristics vary by timing (systolic/diastolic) and valve location
- PhysioNet CinC 2016 Heart Sound Dataset
- Digitized phonocardiogram (PCG) recordings
- Recordings collected at four auscultation sites:
 1. Aortic (AV)
 2. Mitral (MV)
 3. Pulmonic (PV)
 4. Tricuspid (TV)
- Labeled as Normal or Murmur
- Includes real-world noise and variability, making detection challenging
- Patient Features: weight, height, gender, and age

PAST WORKS FOR AUTOMATED HEART MURMUR DETECTION

Traditional Methods

AlBazar etc. (IJA CS & Apps)

- Extracted statistical and frequency-domain features from PCG signals
- Used SVM, Random Forest, AdaBoost, and Decision Trees
- Achieved 85–89% accuracy
- Computationally efficient and interpretable
- Performance depends heavily on handcrafted features

Classical signal processing approaches

- Time- and frequency-domain descriptors (energy, variance, entropy)
- Limited ability to capture complex murmur patterns



Deep Learning

Prince etc. (JAHA, 2023)

- CNN trained on real-world digital stethoscope data
- Performance comparable to trained cardiologists
- Focused on overall murmur detection, not per-valve analysis

Hadad (MSc Thesis, 2025)

- Used spectrograms, wavelet transforms, and Mel features
- CNN and ResNet-18 architectures
- Achieved above 90% accuracy
- Showed benefit of richer time–frequency representations

Alkhodari etc. (IEEE JBHI)

- Used wavelet-based features and attention transformers
- Emphasized valve-specific murmur detection
- Demonstrated strong performance and anatomical relevance

METHODOLOGY: TWO APPROACHES

1D Time-Domain CNN

- Operates directly on filtered PCG waveforms
- Preserves fine temporal structure of heart sounds
- Learns patterns related to S1, S2, systole, and diastole
- Lower computational cost

2D Image Processing Spectrogram CNN

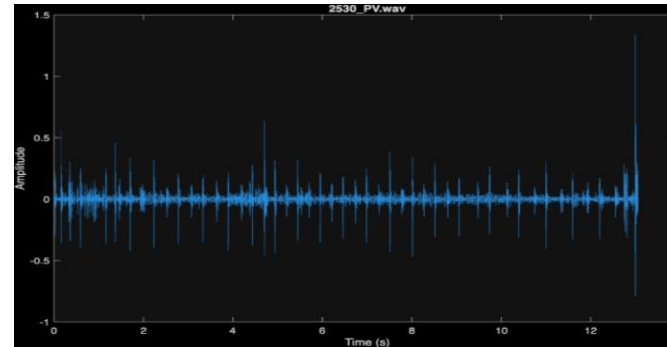
- Converts PCG signals into time–frequency spectrogram images
- Captures both temporal and spectral murmur characteristics
- Great for detecting turbulent flow patterns
- Higher computational cost

Model Performance: confusion matrix and per valve analysis by focusing on sensitivity and specificity.

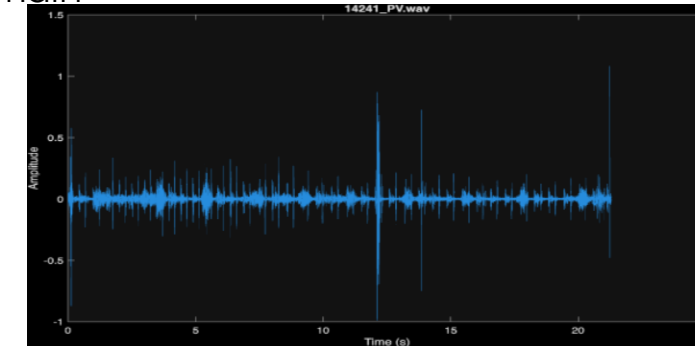
PREPROCESSING PCG SIGNALS

- **Preprocessing of PCG Signals**
- **Bandpass Filtering (20–800 Hz)**
 - Suppresses environmental noise
 - Preserves dominant heart sound and murmur energy
- **Butterworth Filter + Zero-Phase Filtering**
 - Flat passband minimizes signal distortion
 - Zero-phase filtering preserves timing of S1 and S2
- **Class Balancing On Training Set**
 - Murmur and No Murmur classes balanced to reduce bias
 - Validation and test sets left unbalanced to reflect real screening conditions

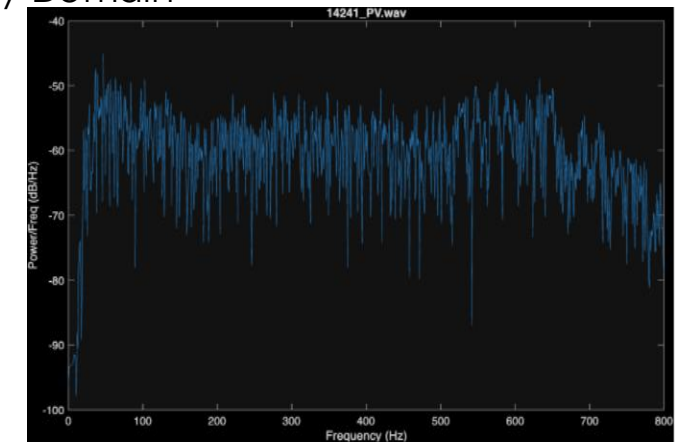
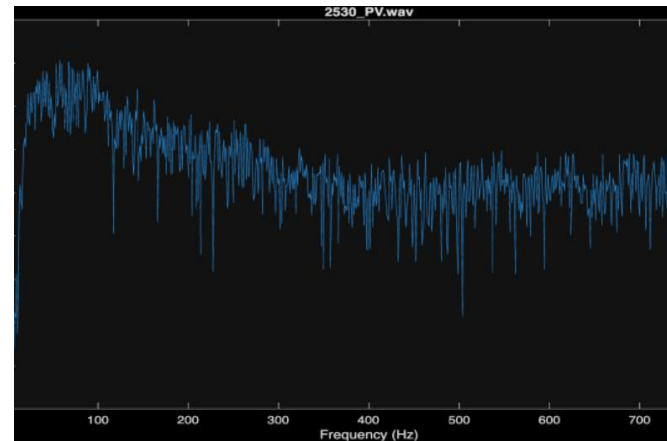
NO MURMUR



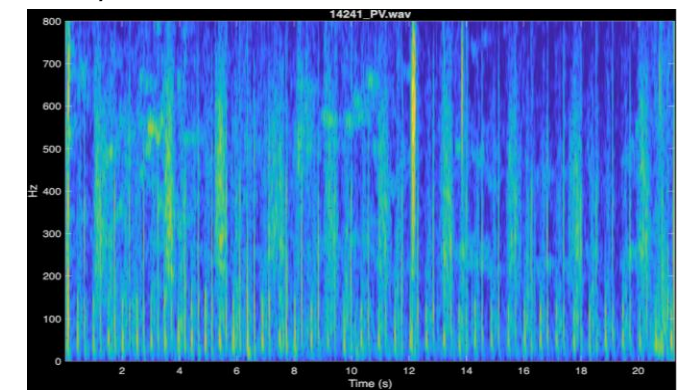
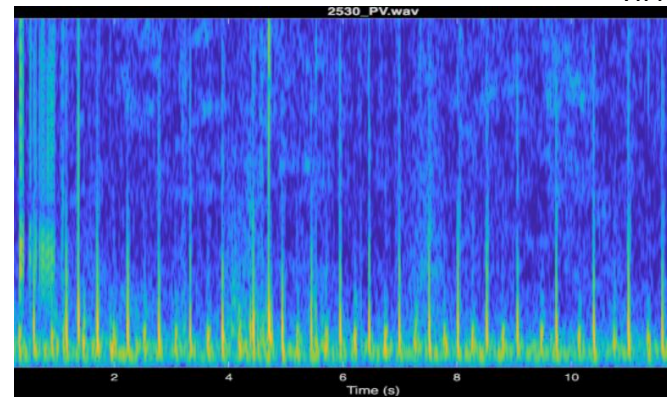
MURMUR



Frequency Domain



Time-Frequency



MODEL-SPECIFIC PREPROCESSING

1D Time Domain CNN

- Resampled to 2000 Hz
- Zero-padding or trimming applied
- Signal normalization (zero mean, unit scale)

2D Spectrogram CNN

- Short-Time Fourier Transform (STFT)
- Frequency range limited to 20–800 Hz
- Resized to 224×224 for CNN input

TRAINING MODELS

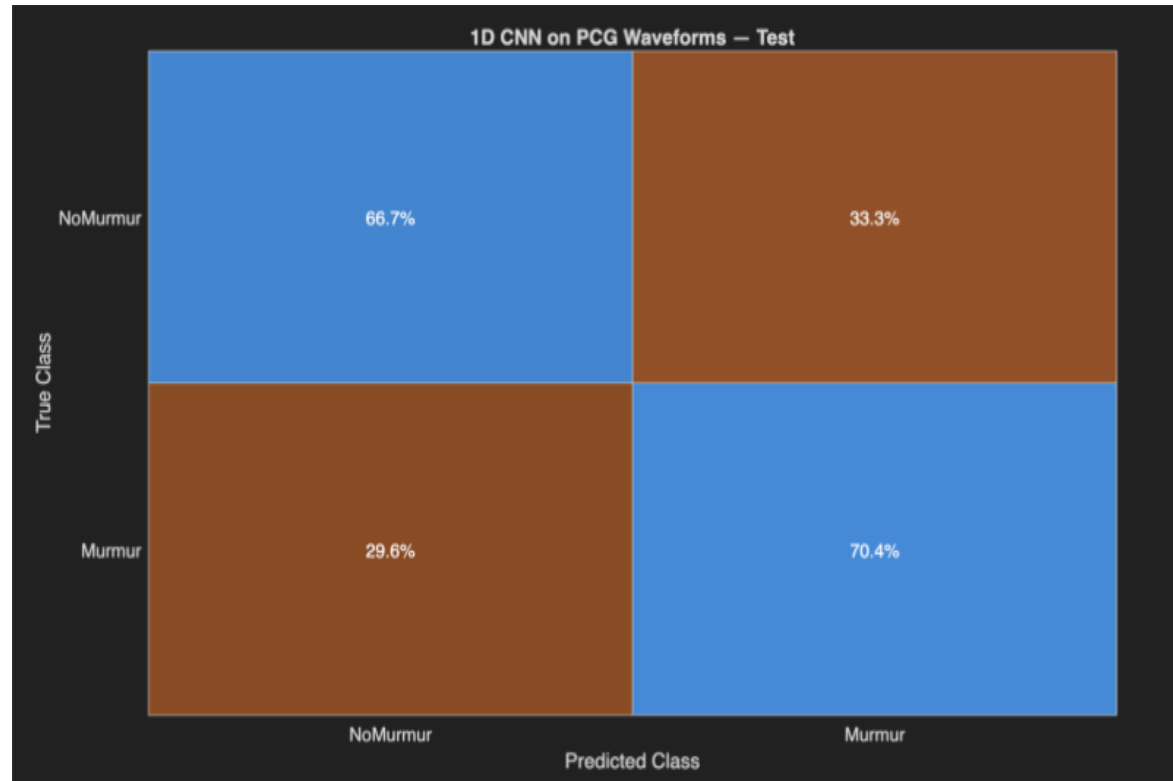
1D Time-Domain CNN

- Input: fixed-length PCG waveforms
- Three 1D convolutional layers with increasing filter depth
- Wider kernels used to capture systolic and diastolic timing patterns
- Each layer includes ReLU activation, batch normalization, and max pooling
- Dropout applied in deeper layers to reduce overfitting
- Final fully connected layer with softmax output
- Adam Optimizer
- Learning rate: 3×10^{-4}
- Batch size: 64

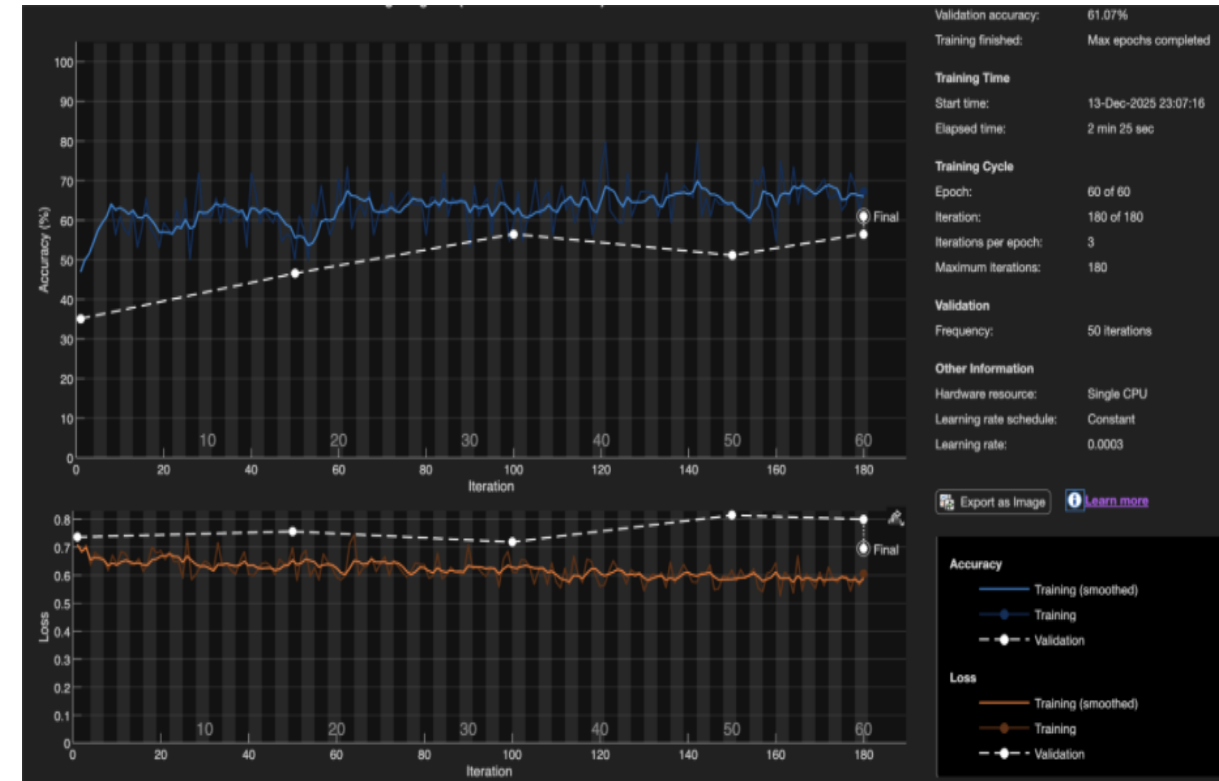
2D Image Processing Spectrogram CNN

- Input: time–frequency spectrogram images
- Three 2D convolutional layers with increasing filter depth
- Learns low-level spectral edges, mid-level frequency patterns, murmur structures
- Each layer includes ReLU, batch normalization, and max pooling
- Dropout and global average pooling used for better generalization
- Final classification layer with softmax output
- Adam Optimizer
- Learning rate: 1×10^{-3}
- Batch size: 32

RESULTS: 1D TIME-DOMAIN CNN

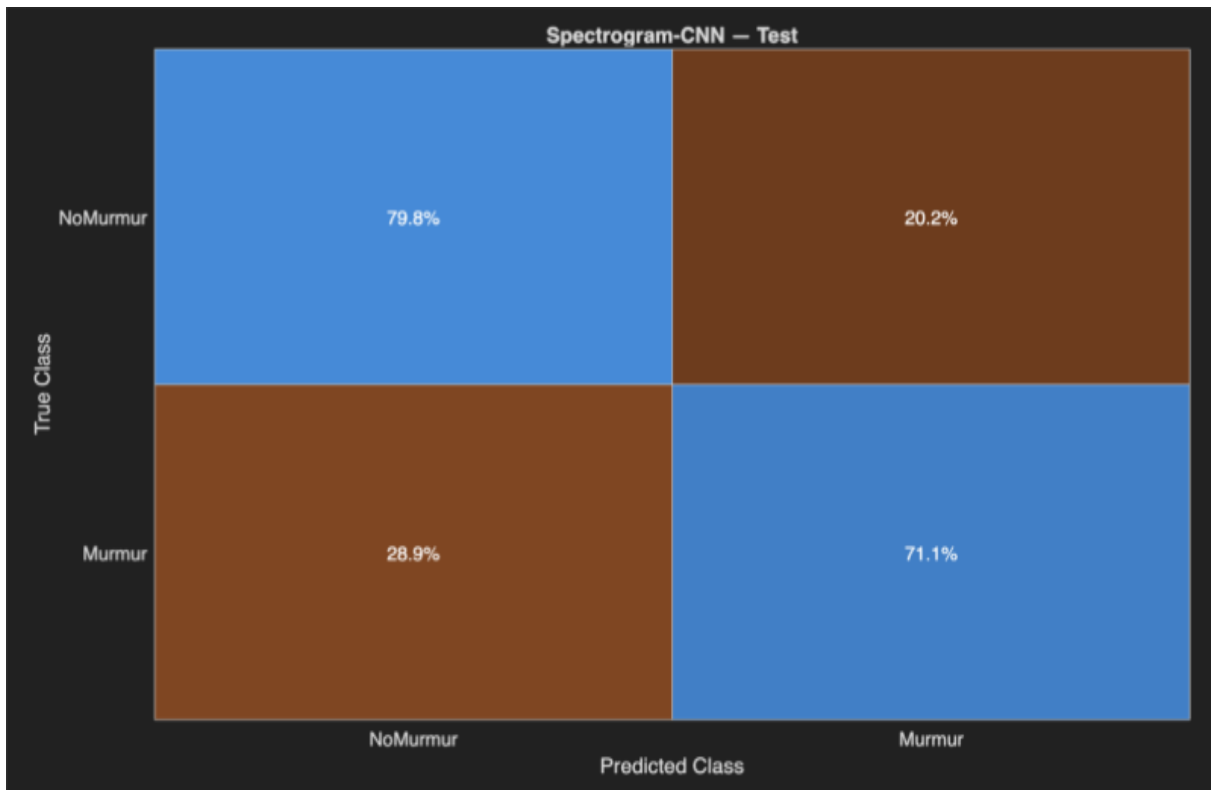


One-Dimensional Convolutional Neural Network
on PCG Waveforms Confusion Matrix

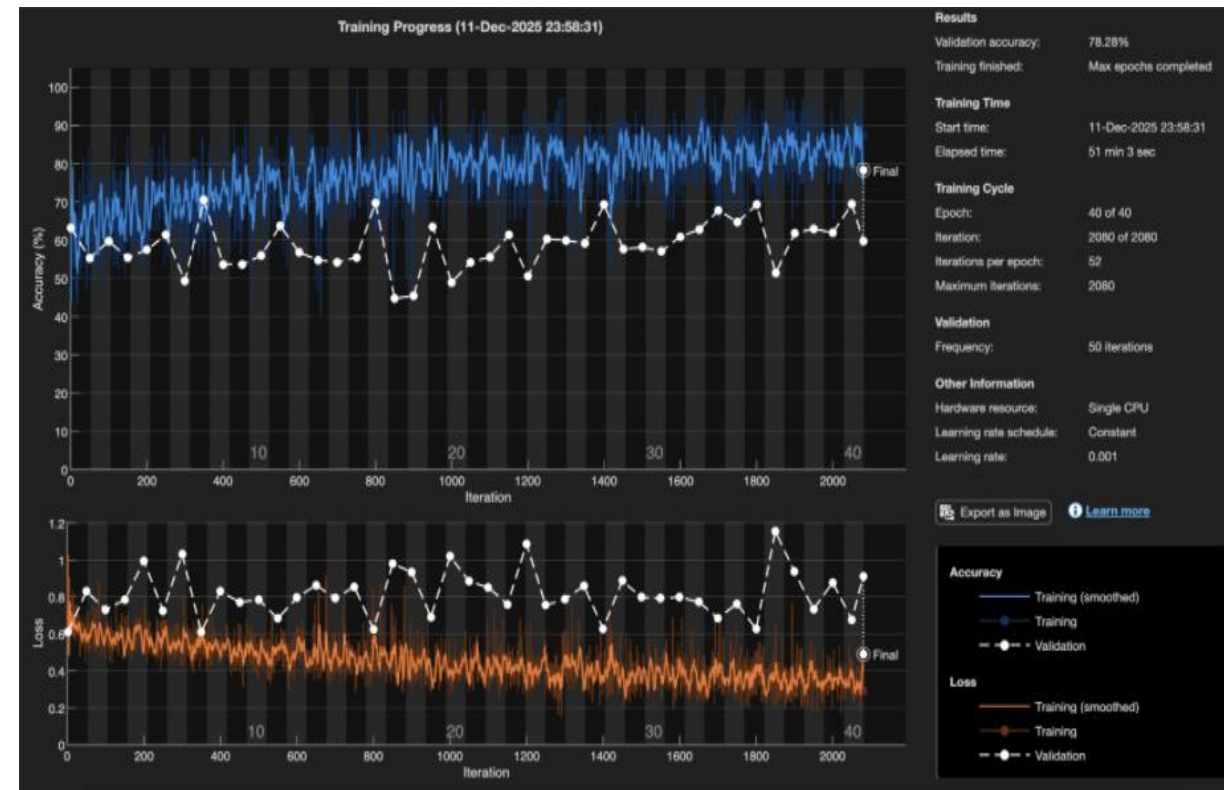


Training and validation accuracy loss curves for
the One-Dimensional Time-Domain Convolution
Neural Network

RESULTS: 2D IMAGINE PROCESSING SPECTROGRAM CNN



Two-Dimensional Spectrogram
Convolution Neural Network Confusion
Matrix



Training and validation accuracy loss curves for
the 2D spectrogram Convolutional Neural
Network

Valve Sites	Model	Murmurs Detected (%)	No-Murmurs Detected (%)
Aortic (AV)	1D Time-Domain	69.60%	66.70%
	2D Spectrogram	73.90%	78.30%
Mitral (MV)	1D Time-Domain		
	2D Spectrogram	76.90%	85.70%
Pulmonic (PV)	1D Time-Domain	50.00%	71.40%
	2D Spectrogram	70.00%	72.10%
Tricuspid (TV)	1D time-Domain		
	2D Spectrogram	61.90%	83.30%

PER VALVE ANALYSIS: *MODEL COMPARISON*

CONCLUDING REMARKS

Discussion

- Both models were able to detect heart murmurs, showing that clinically relevant information exists in both the time domain and time–frequency domain
- The 1D waveform CNN showed lower specificity and inconsistent performance across valves
- The 2D spectrogram CNN produced more stable and clinically reliable results by capturing both temporal and spectral features
- Per-valve analysis showed the aortic and mitral valves were the most informative sites, while limited data reduced reliability at other locations
- Overall results highlight a tradeoff between computational efficiency and classification accuracy

Future Work

- Increase dataset size to improve generalization and per-valve reliability
- Explore multichannel recordings to jointly learn relationships between valve sites
- Investigate hybrid models (CNN + attention/transformers or wavelet-based features)
- Improve low-cost 1D models for resource-constrained clinical settings
- Extend classification beyond binary detection to valve-specific murmur types

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Questions ?

THANK YOU!

